# Linear Approximation and L'Hôpital's Rule

**ANSWER KEY**

## Section 1: Tangent Line Approximation (Linearization)

**1** What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

**a.** $L(x) = 5 + \frac{1}{10}(x - 25)$ ✓

**b.** $L(x) = 5 + \frac{1}{5}(x - 25)$

**c.** $L(x) = 5 + \frac{1}{2}(x - 25)$

**d.** $L(x) = \frac{1}{10}x$

$f(25) = 5$. $f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(25) = \frac{1}{10}$. Linearization: $L(x) = f(a) + f'(a)(x - a) = 5 + \frac{1}{10}(x - 25)$.

**2** Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

**a.** 5.10 ✓

**b.** 5.05

**c.** 5.20

**d.** 5.01

$L(26) = 5 + 0.1(26 - 25) = 5 + 0.1 = 5.10$.

**3** Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?

**a.** **An overestimate because $f''(x) < 0$ (concave down)** ✓

**b.** An underestimate because $f''(x) > 0$ (concave up)

**c.** Exact because $f$ is monotonic

**d.** Depends on the step size

Since $f''(x) = -\frac{1}{4}x^{-3/2} < 0$, the curve is concave down and lies below its tangent lines, making tangent line approximations overestimates.

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## Section 2: L'Hôpital's Rule for 0/0 Forms

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**4** Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{\sin(3x)}$.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{3}{2}$

**c.** 0

**d.** $\infty$

Direct substitution yields $\frac{0}{0}$. By L'Hôpital's Rule: $\lim_{x \to 0} \frac{2e^{2x}}{3\cos(3x)} = \frac{2(1)}{3(1)} = \frac{2}{3}$.

**5** Evaluate $\lim_{x \to 1} \frac{\ln x}{x^2 - 1}$.

**a.** $\frac{1}{2}$ ✓

**b.** 1

**c.** 2

**d.** 0

$\frac{0}{0}$ form. By L'Hôpital's: $\lim_{x \to 1} \frac{1/x}{2x} = \frac{1}{2(1)} = \frac{1}{2}$.

**6** Evaluate $\lim_{x \to 0} \frac{1 - \cos(4x)}{x^2}$.

**a.** 8 ✓

**b.** 4

**c.** 2

**d.** 16

First L'Hôpital's: $\frac{4\sin(4x)}{2x}$. Second L'Hôpital's: $\frac{16\cos(4x)}{2} = \frac{16(1)}{2} = 8$.

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## Section 3: L'Hôpital's Rule for Infinity/Infinity Forms

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**7** Evaluate $\lim_{x\to\infty} \frac{\ln(x)}{x}$.

**a.** 0 ✓

**b.** 1

**c.** $\infty$

**d.** Undefined

$\frac{\infty}{\infty}$ form. By L'Hôpital's: $\lim_{x\to\infty} \frac{1/x}{1} = 0$.

**8** Evaluate $\lim_{x\to\infty} \frac{x^2}{e^x}$.

**a.** 0 ✓

**b.** 1

**c.** 2

**d.** $\infty$

Apply L'Hôpital's twice: $\lim_{x\to\infty} \frac{2x}{e^x} = \lim_{x\to\infty} \frac{2}{e^x} = 0$.

**9** Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x\to c} \frac{f(x)}{g(x)}$?

**a.** **The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$** ✓

**b.** $f(c) = 0$ only

**c.** $f'(c) = 0$

**d.** $g(x) > 0$ for all $x$

L'Hôpital's Rule applies ONLY when the quotient evaluates to an indeterminate form of type $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$.

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## Section 4: Comparing Relative Growth Rates of Functions

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**10** As $x \to \infty$, which of the following functions grows the fastest?

**a.** $e^x$ ✓

**b.** $x^{10}$

**c.** $x \ln x$

**d.** $2^x$

Exponentials with larger base grow faster than any polynomial or power function. Since $e \approx 2.718 > 2$, $e^x$ grows fastest.

**11** Evaluate $\lim_{x \to \infty} \frac{(\ln x)^5}{\sqrt{x}}$.

**a.** $0$ ✓

**b.** $\infty$

**c.** $1$

**d.** $5$

Any power of $x$ (even $x^{1/2}$) grows strictly faster than any power of $\ln x$. Hence the limit is $0$.

**12** If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

**a.** $f(x)$ **grows faster than** $g(x)$ ✓

**b.** $g(x)$ grows faster than $f(x)$

**c.** They grow at the same rate

**d.** Neither function grows

By definition of relative growth rates, $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$ indicates that $f(x)$ grows strictly faster than $g(x)$.

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